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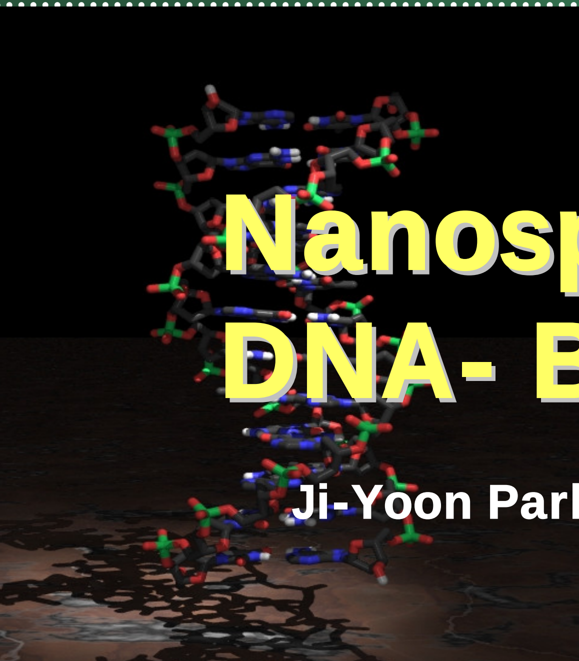

Nanospheres for DNA- Based Theorem Proving

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Preview

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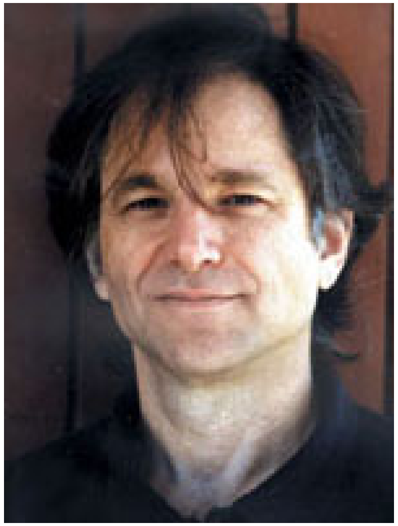
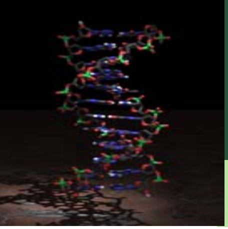
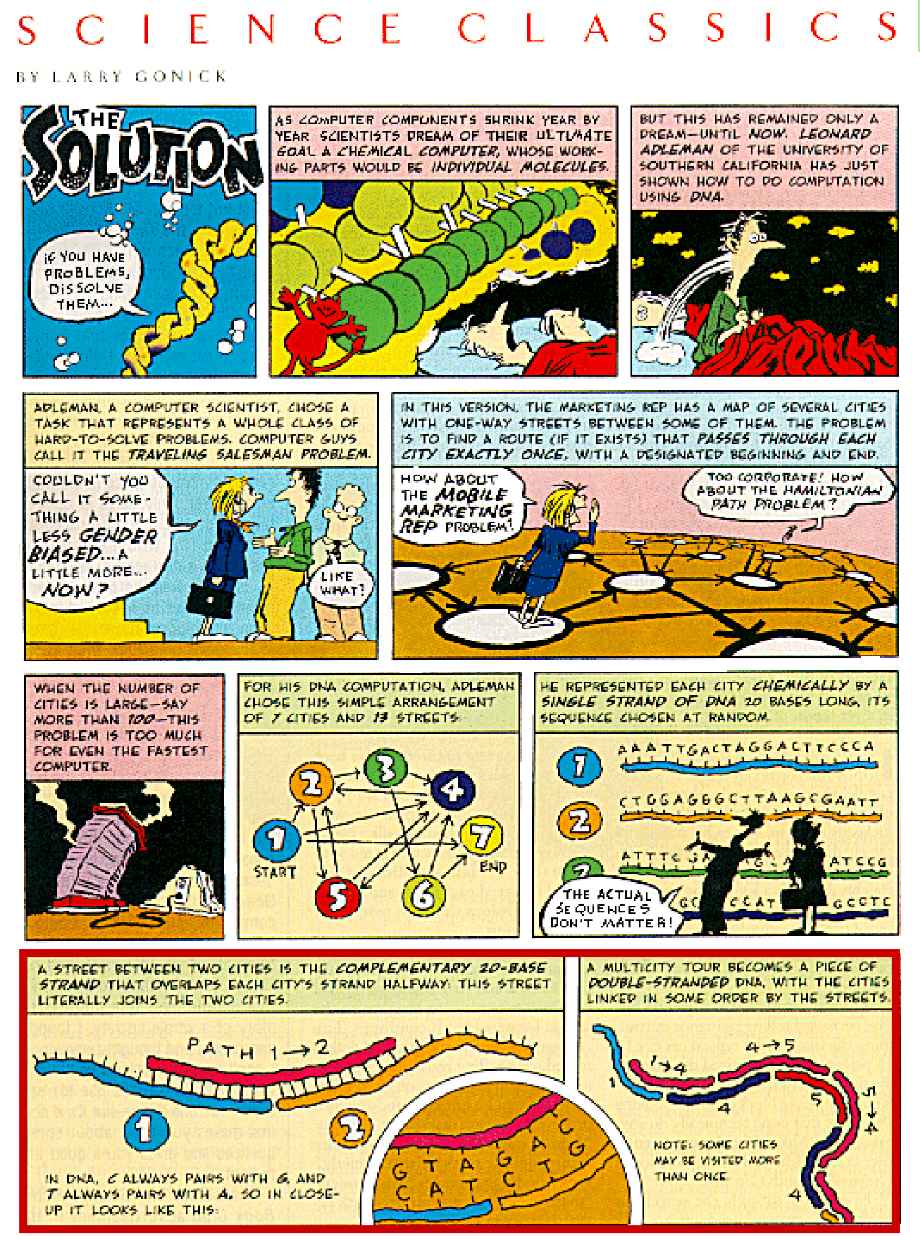
I believe things like DNA computing will eventually lead the way to a "molecular revolution" which ultimately will have a very dramatic effect on the world.

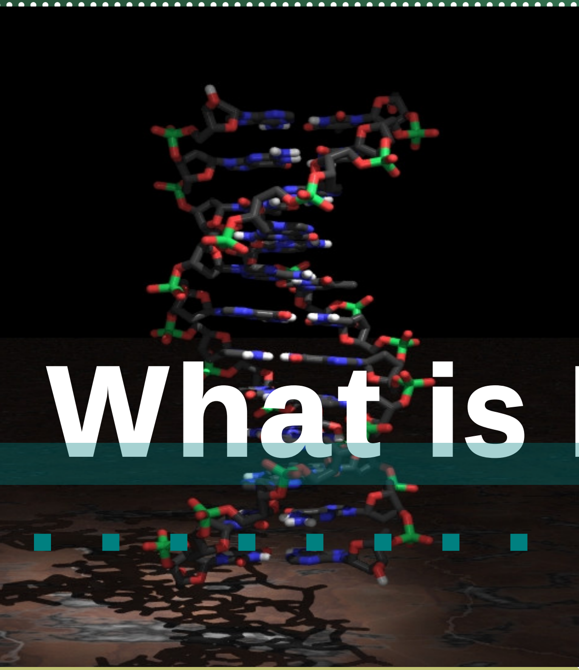
– L. Adleman



Adleman's Idea



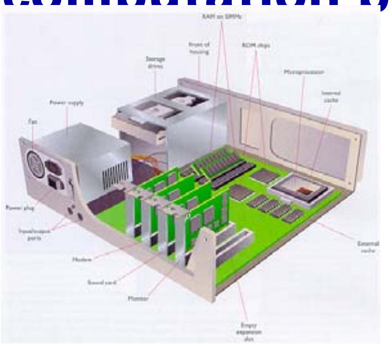




What is DNA Computing?

What is DNA Computing?

The use of biological molecules, primarily DNA, DNA analogs, and RNA, for computation purpose.



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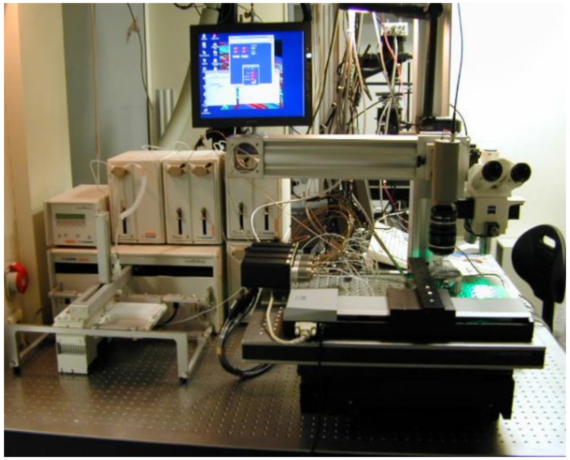
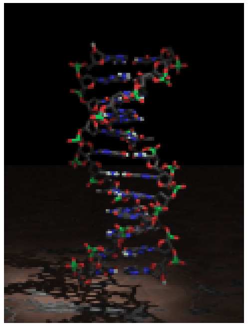




ATGCTCGAAGCT

Historical Timeline

Research

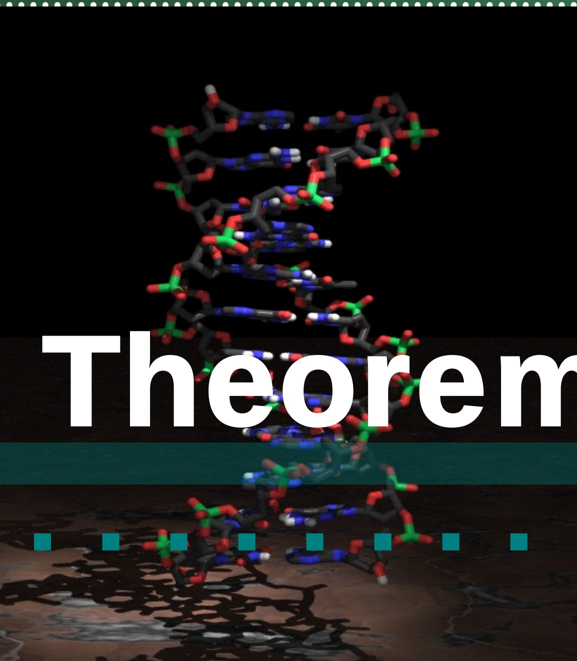
1961	1994	1995	2004	2005
R.Feynman's paper on sub microscopic computers	L.Adleman solves Hamiltonian path problem using DNA.Field started	D.Boneh paper on breaking DES with DNA	Shapiro's paper on logical control of gene expression	DNA computer architecture ?

Commercial

1970's ...	1996	2000	2015
DNA used in bio application	Affymetrix sells GeneChip DNA analyzer	Human Genome Sequence	Commercial computer ?

DNA Computers vs. Conventional computers

DNA-based computers	Microchip-based computers
Can do billions of operations simultaneously **(massive parallelism)**	Can do substantially fewer Operations simultaneously
Can provide **huge memory** in small space	Smaller memory
Setting up a problem may involve considerable preparations	Setting up only requires keyboard input
DNA is sensitive to chemical deterioration	Electronic data are vulnerable but can be backed up easily



Theorem Proving

Theorem Proving: Historical Perspective

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- Theorem proving (or automated deduction)
  = logical deduction performed by machine
  = A method used for automated reasoning
  = Given rules and facts, machine draws new conclusion by itself using inference rules

- **At the intersection of several areas**
  - Mathematics: original motivation and techniques
  - Logic: the framework and the meta-reasoning techniques

- **One of the most advanced and technically deep fields of computer science**
  - Some results as much as 75 years old
  - Automation efforts are about 40 years old

Previous Research

- **Hairpin DNA was used for horn clause computation**
  [Uejima et al, 2001]

- **Linear DNA in various applications related with resolution**
  - Inference system [Wasiewicz *et al*., 2000]
  - Horn clause computation [Kobayashi, 1999]
  - PROLOG interpreter [Mihalache, 1997]

Applications

- **Software / hardware productivity tools**
  - Hardware and software verification (or debugging)
  - Security protocol checking

- **Automatic program synthesis from specifications**

- **Discovery of proofs of conjectures**
  - A conjecture of Tarski was proved by machine (1996)
  - There are effective geometry theorem provers

Resolution Principle(1)

- Resolution *refutation* proves a theorem by negating the statement to be proved and adding <u>this negated goal</u> to the set of axioms that are known to be true.

- Use the resolution rule of inference to show that this leads to a contradiction.

- Once the theorem prover shows that the negated goal is inconsistent with the given set of axioms, it follows that the original goal must be consistent.

Resolution Principle(2)

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- **Steps for resolution refutation proofs**
  - Put the premises or axioms into clause form.
  - Add the negation of what is to be proved, in clause form, to the set of axioms.
  - Resolve these clauses together, producing new clauses that logically follow from them.
  - Produce a contradiction by generating the empty clause.
  - The substitutions used to produce the empty clause are those under which the opposite of the negated goal is true.

List of Symbols

Symbol	Meaning
$\neg P$	not P
$P \wedge Q$	P and Q
$P \vee Q$	P or Q
$P \rightarrow Q$	if P then Q
	$\neg P \vee Q$

Resolution Refutation (1/2)

If someone has a car$_C$ and an account with a bank$_A$, he (she) have a good credit$_R$.

$C \wedge A \rightarrow R$

If someone owns a house$_H$, then he (she) has an account with a bank$_A$.

$H \rightarrow A$

Someone owns a house$_H$ and has a car$_C$. Is he (she) have a good credit$_R$?

$H, C,$ R is True ?

$C \wedge A \rightarrow R,\ H \rightarrow A,\ H,\ C,$ R is True ?

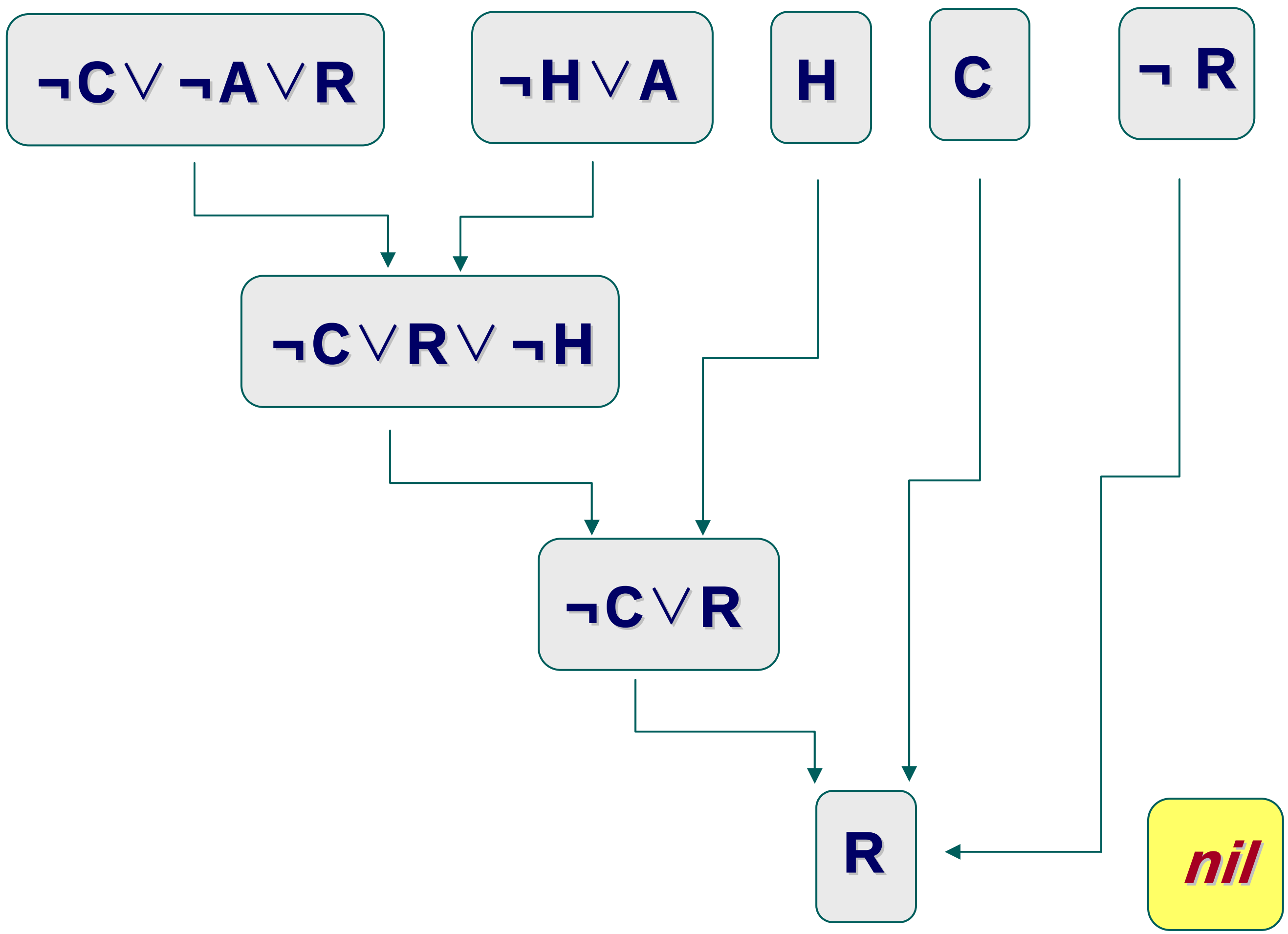
$\Downarrow$

$\boxed{(\neg C \vee \neg A \vee R)} \wedge (\neg H \vee A) \wedge H \wedge C \wedge \boxed{\neg R}$

A Clause

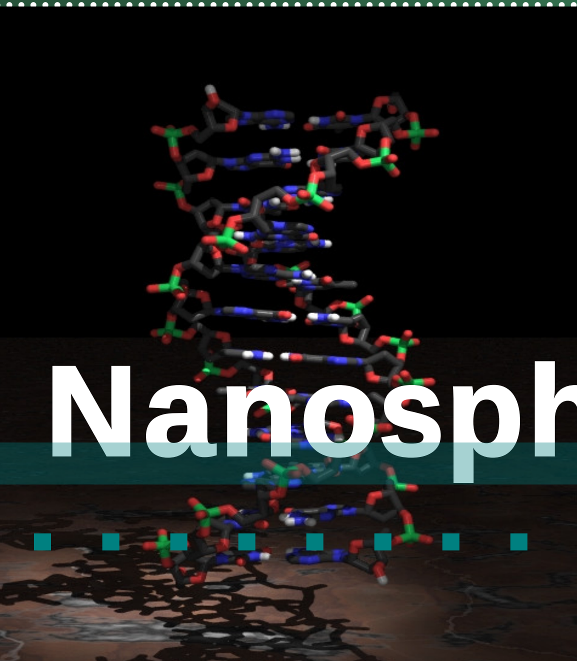
Resolution Refutation (2/2)

$$(\neg C \lor \neg A \lor R) \land (\neg H \lor A) \land H \land C \land \neg R$$



Purpose

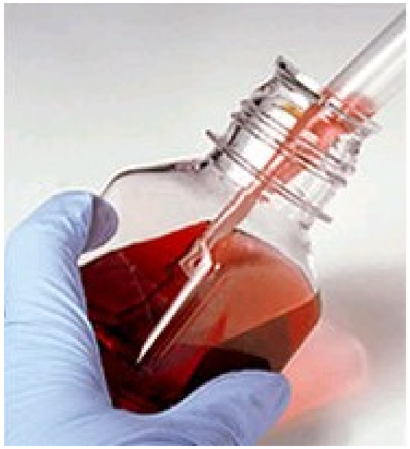
- To prove theorem proving using method with rational assembling nanoparticles



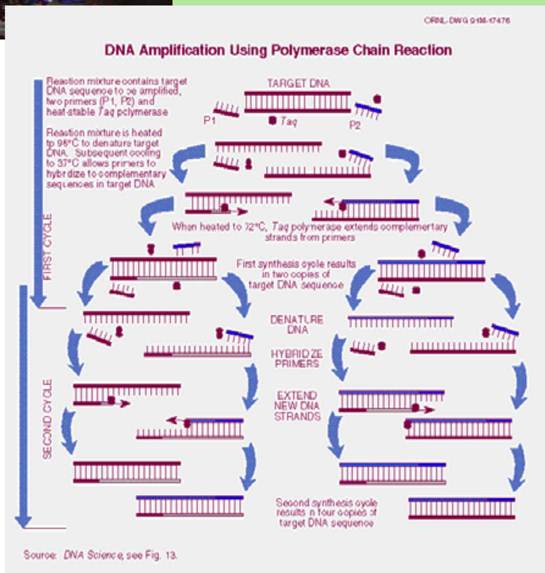
Nanospheres

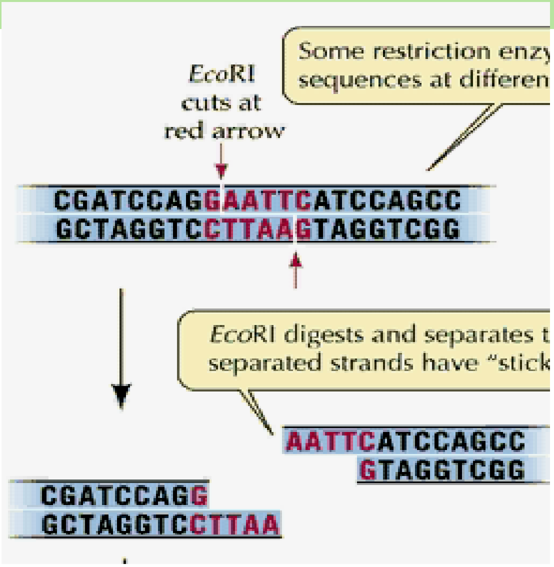
Why Gold Nanoparticle ?

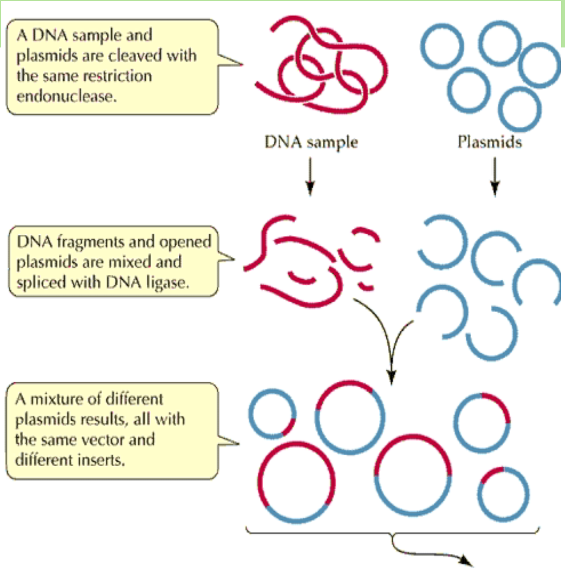
- **Sensitivity**
  - The perfect material for rapid testing
  - Visual signal
- **Stability**
- **Surface Area**
  - Intrinsic spherical form
- **Application**
  - Small size & increased surface area
  - DNA, protein, enzyme...

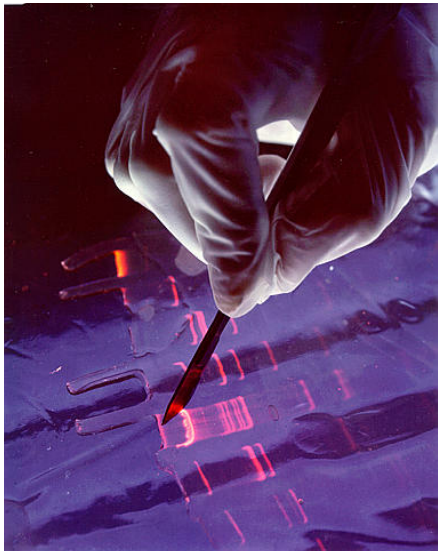


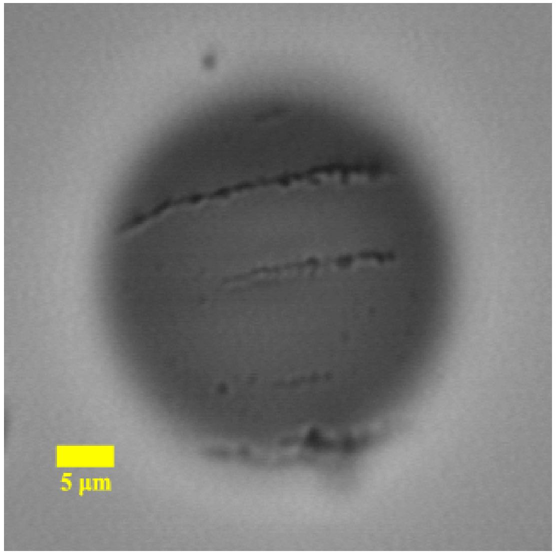
Previous Operator



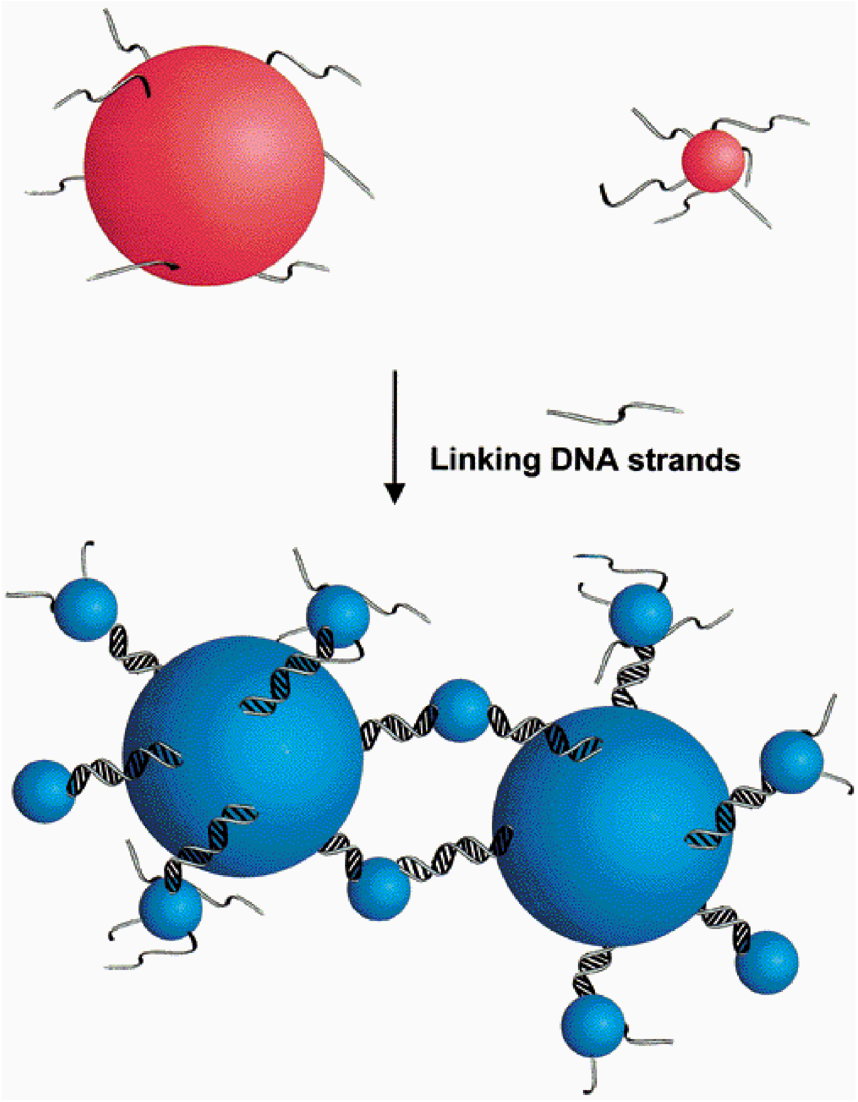


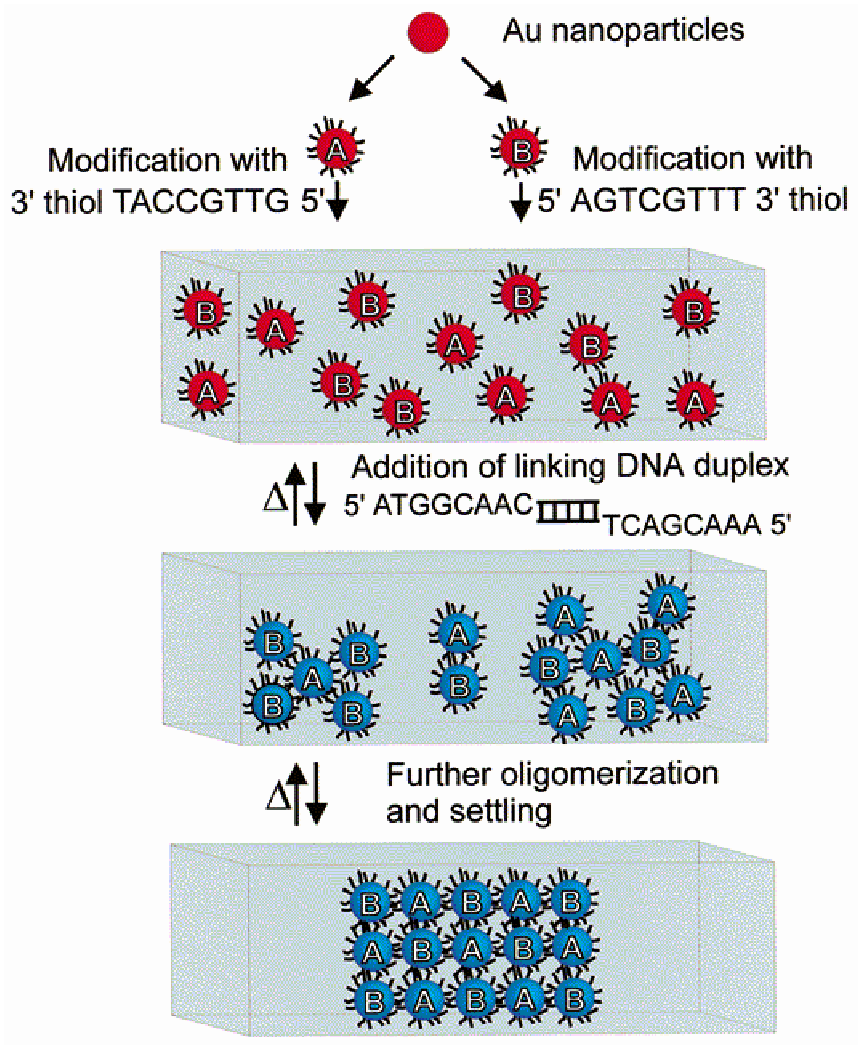






DNA-based Nanoparticle Assembly





Nature, 382:607-609

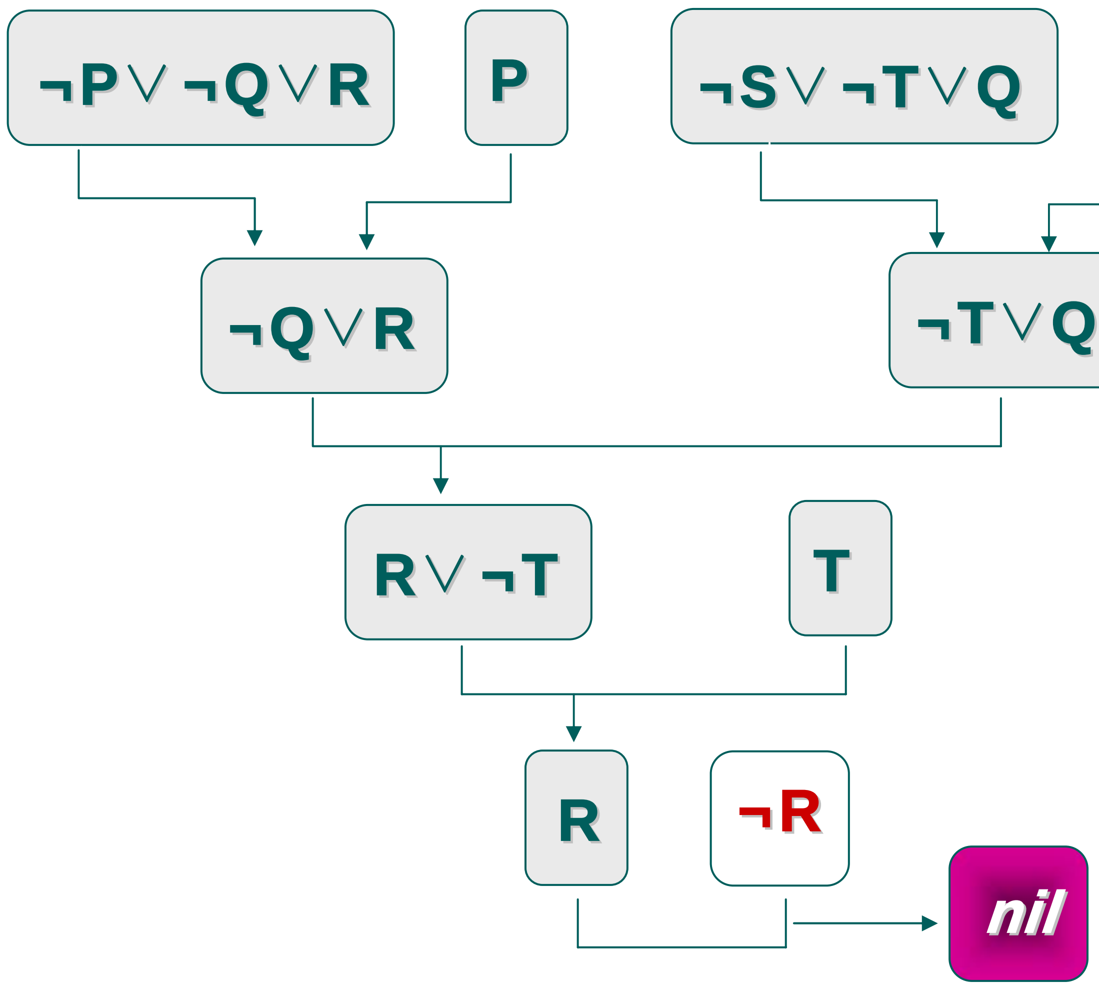
Target Problem

$P \wedge Q \rightarrow R$, $S \wedge T \rightarrow Q$, S, T, P

R = True ?

Resolution Refutation

$(\neg P \vee \neg Q \vee R) \wedge (\neg S \vee \neg T \vee Q) \wedge S \wedge T \wedge P \wedge \neg R$



Encoding of DNA Molecules

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- **Sequence for literals in one clause**
- **Simple method**

A → 5'-ACGTTAGA-3'

¬A → 5'-TCTAACGT-3'

B → 5'-TCGTCAGC-3'

¬B → 5'-GCTGACGA-3'

Two types of Assembly

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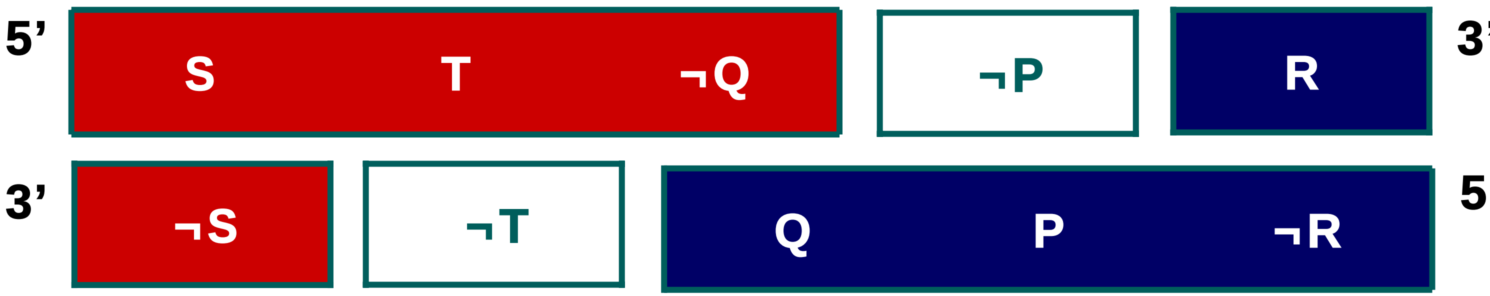
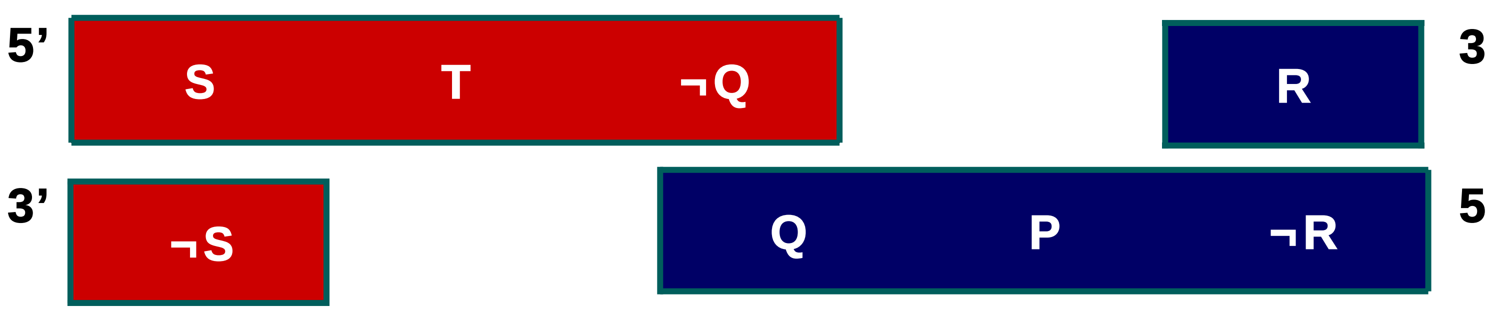
(a) Full assembly



(b) Partial assembly



Experimental Section

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1. Preparation of gold nanoparticles

↓

2. Synthesis of (alkanthiol)-modified oligonucleotides

↓

3. Preparation of 3'-(alkanthiol)oligonucleotides-modified gold nanoparticles

↓

4. Preparation of linker DNA

↓

5. Assembly & S1 Nuclease digestion

↓

6. Characterization of assembly reaction

Color Change of DNA-Induced Assembly

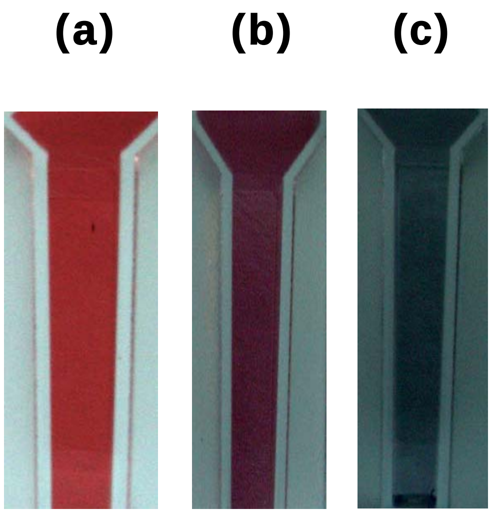

Figure 1. Color change of DNA-linked assembly

(a) The 10-nm gold particles

(b) The Solutions of DNA-linked full assembly

(c) Aqueous solution of the addition of NaCl

UV-Vis Spectrum of 10-nm Gold Particles

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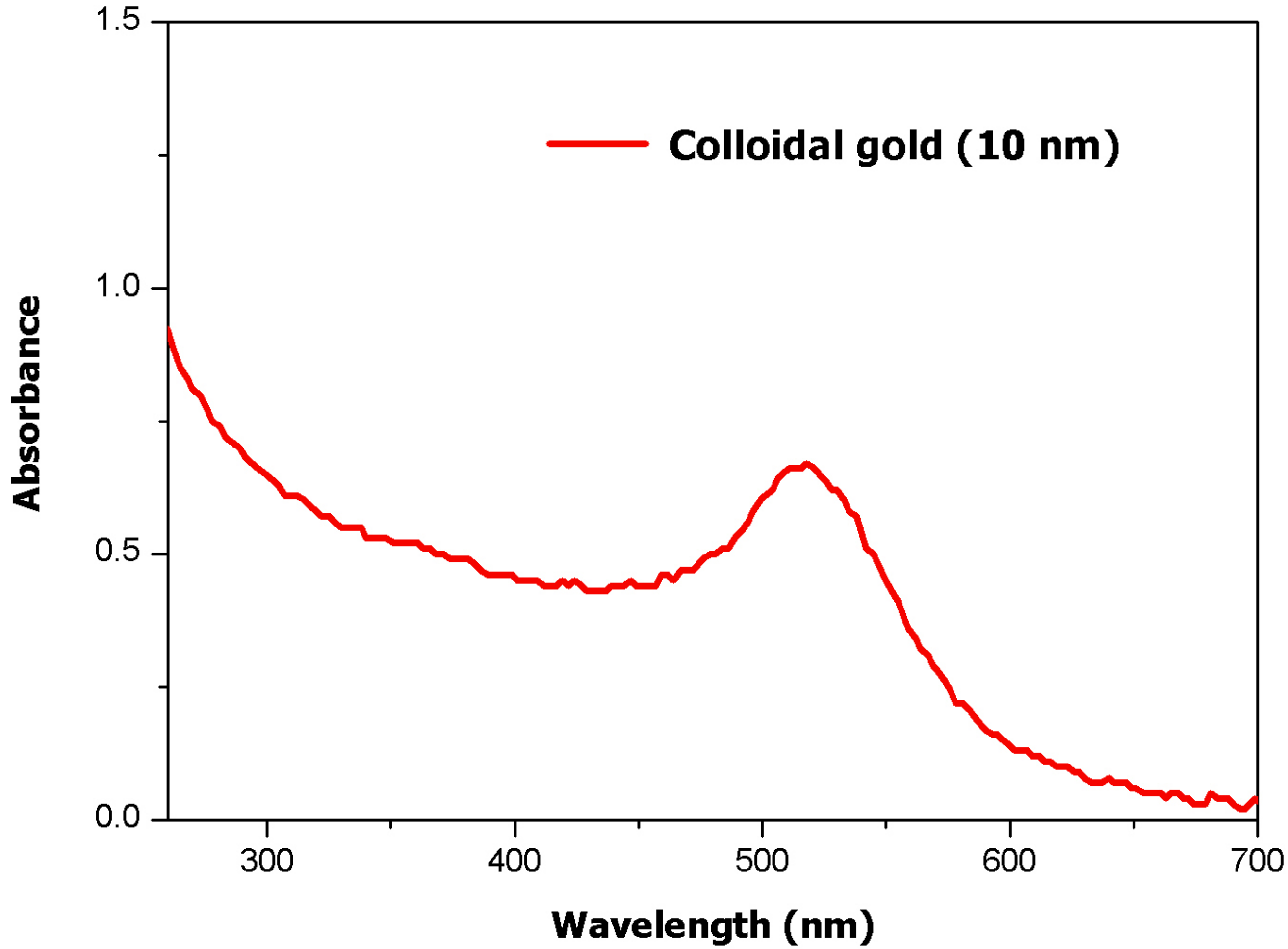
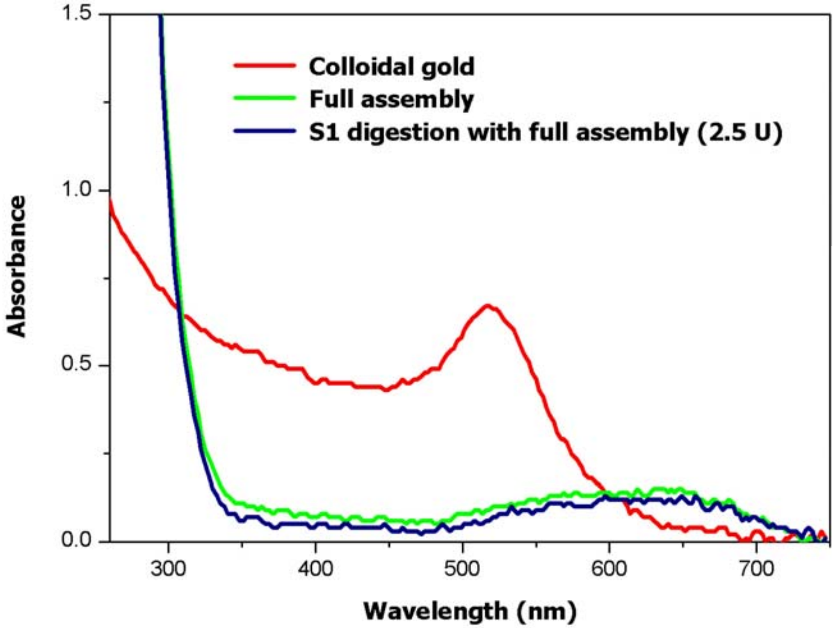
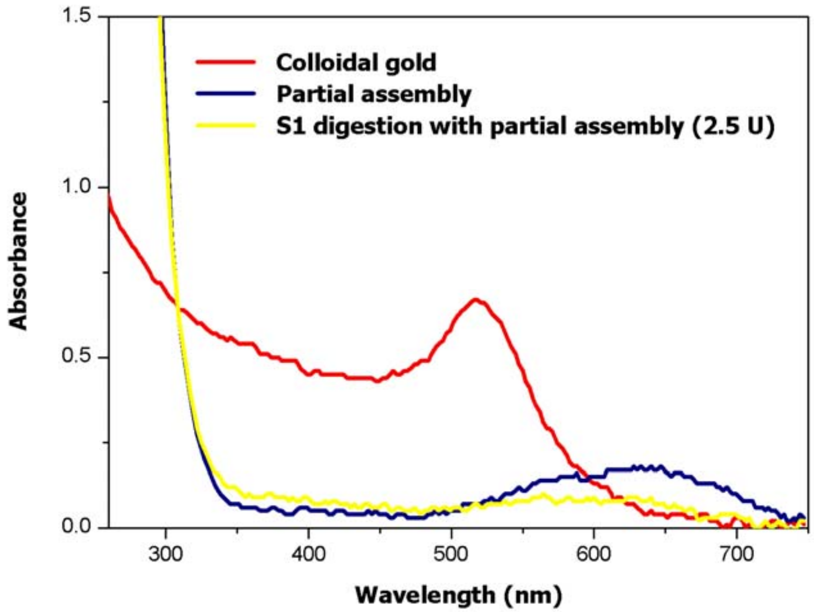


UV-Vis Spectra of Assembly and S1 Digestion

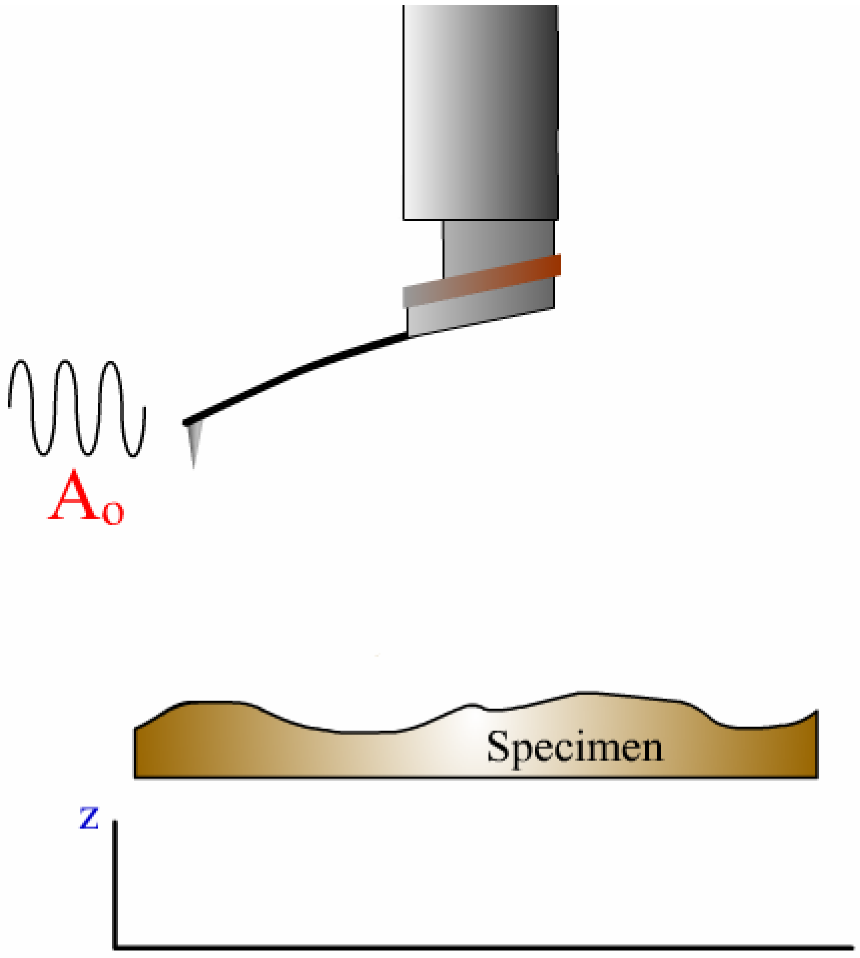
Full Assembly vs S1 Nuclease Digestion



Partial Assembly vs S1 Nuclease Digestion



Atomic Force Microscope (AFM)



 www.ntmdt.com

AFM Analysis of DNA-Linked Full Assemb

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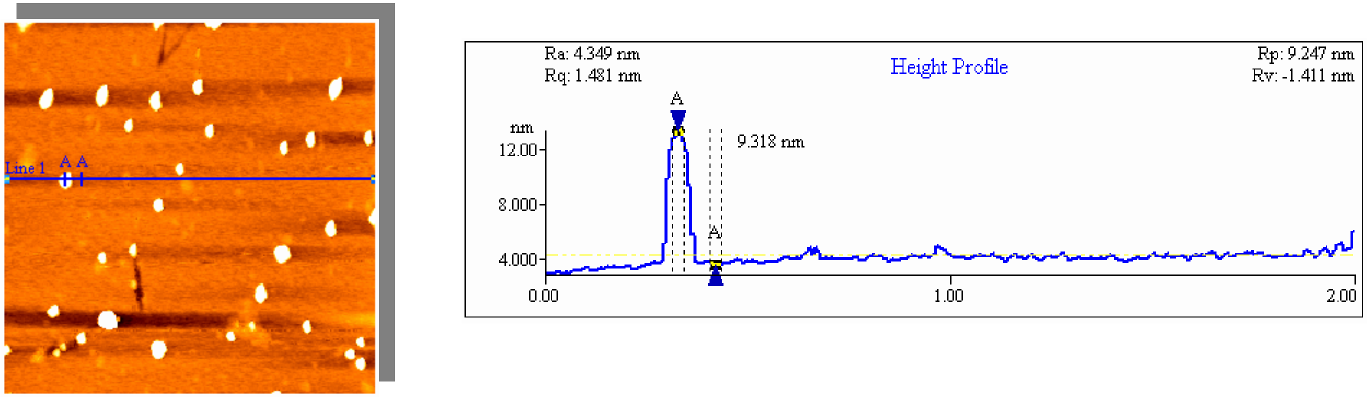
(a)



(b)

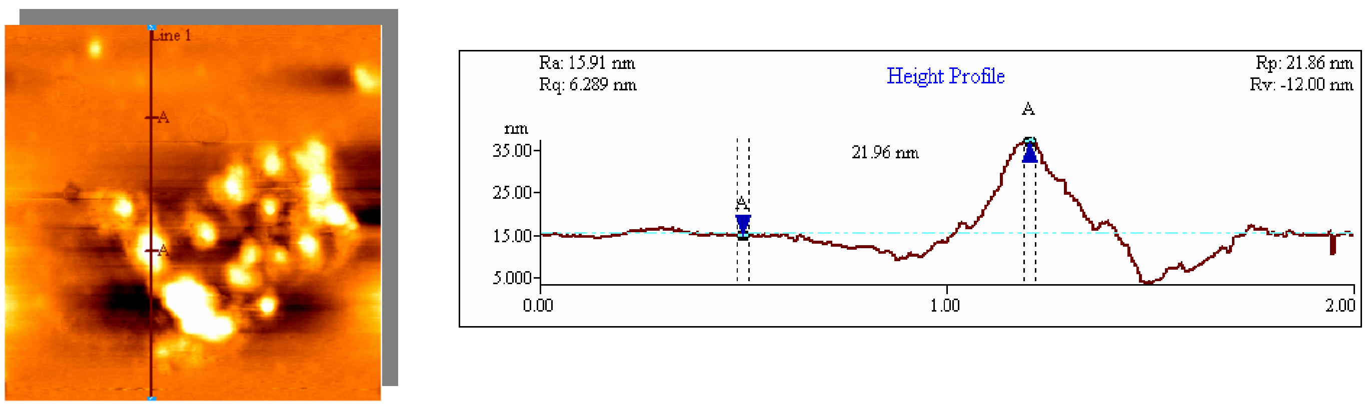


Figure 2. AFM Analysis of DNA-linked full assembly

(a) The 10-nm gold nanoparticles

(b) DNA-linked full assembly

Transmission Electron Microscopy (TEM)

Transmission Electron Microscope (TEM) Image of Full Assembly

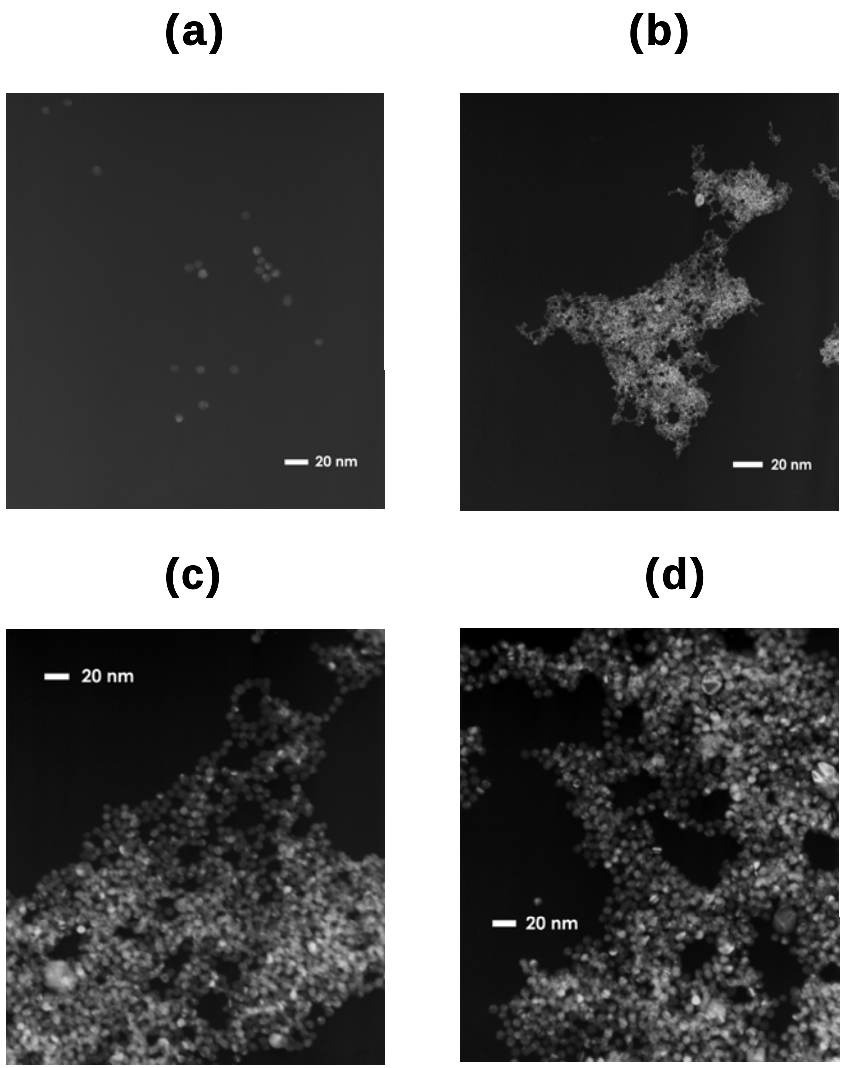


Figure 3. TEM image of full assembly

(a) Image of 10-nm gold particles

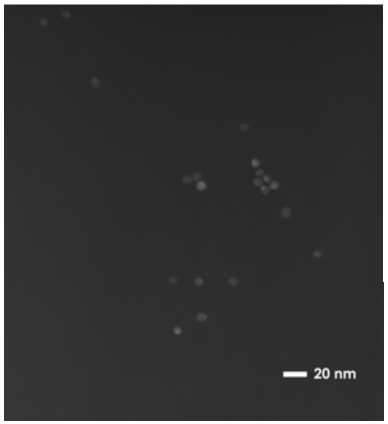
(b) DNA-linked full assembly

(c) A higher magnification image (x 200 k) of a portion of the assembly displayed in panel (a)

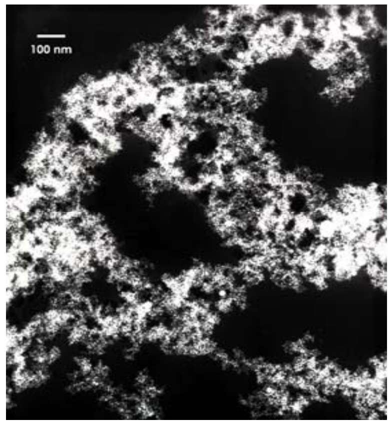
(d) S1 digestion of full assembly

Transmission Electron Microscope (TEM) image of Partial Assembly

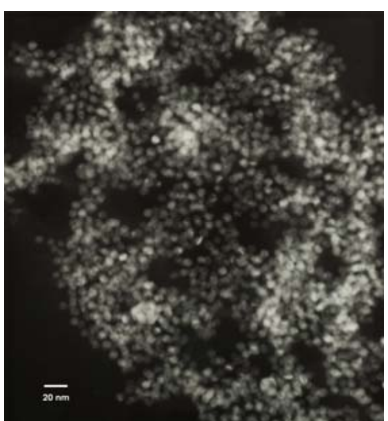
(a)



(b)



(c)



(d)

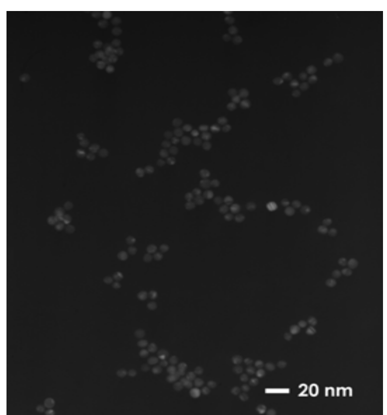


Figure 4. TEM image of partial assembly

(a) Image of 10-nm gold particles

(b) DNA-linked partial assembly

(c) A higher magnification image (x 200 k) of a portion of the assembly displayed in panel (a)

(d) S1 digestion of partial assembly

Summary

- We demonstrated an implementation of **nanoparticle-based approach in theorem proving.**

- We approached **the assembly phenomenon of DNA-functionalized gold nanoparticles** induced by hybridization of target DNA which cross-links to the nanoparticle.

Conclusions

- Finally, we examined **nanoparticle-based theorem proving** by colorimetric detection. Nanoparticle-based theorem proving was proved by UV-Vis spectra, AFM and TEM.

- Furthermore, **the expansion of theorem proving** will be applied to branch DNA structure and DNA tile. For expansion, exquisite optimization of the encoding process is necessary. This experimental system will prove fruitful in further investigations in theorem proving by resolution refutation.

Rapid Detection

Parallel Reaction in Theorem Proving

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